Infant

Irish Centre for Maternal and Child Health Research

CTG Data Analysis and Annotation Web Tool

Software Requirements Specification (SRS)

PRD-Style Introduction

CTG Data Analysis and Annotation Web Tool

Software Requirements Specification (SRS) with PRD-Style Introduction

Non-Technical Overview (Clinical and Business)

What this tool is

This tool helps clinicians and reviewers analyze historical CTG traces using a clear, guided workflow. It highlights important patterns, applies guideline-aligned classifications, and supports manual review before results are finalized.

Who this is for

- Nurses and midwives reviewing CTG traces.
- Obstetric and clinical reviewers supporting second checks.
- Educators and research teams reviewing historical cases.
- Product and operations stakeholders tracking quality and consistency.

Why this matters

- Improves consistency by applying the same rule logic across cases.
- Reduces review friction with a guided Upload -> Analyze -> Review -> Export flow.
- Makes decisions easier to explain with visible rule triggers and evidence.
- Supports safer handovers with exportable summaries and traceable edits.

What this tool does not do

- It does not replace clinical judgment.
- It does not provide treatment decisions.
- It is not an autonomous diagnostic system.
- Final responsibility remains with qualified clinicians.

Typical nurse workflow (plain language)

- Step 1: Upload a CTG file for a historical case.
- Step 2: View FHR and uterine activity traces on one screen.
- Step 3: Run analysis to generate initial labels (Normal, Suspicious, Pathological).
- Step 4: Review highlighted events and explanations.
- Step 5: Accept or edit annotations where needed.
- Step 6: Export a summary for documentation, review, or teaching.

Expected user experience goals

- First-time users can complete a basic review in about 10 minutes.
- Key terms and messages are clear and non-technical where possible.
- Warnings and errors explain what happened and what to do next.
- Color is supported by labels/icons so meaning is not color-only.

Business and operational outcomes

- Faster retrospective case review for audit and education.
- More consistent classification across users and sessions.
- Better traceability of who changed what and when.
- Reusable exports for reporting, QA, and supervisor sign-off.

Pilot success metrics (non-technical)

- Time to complete one standard case review.
- Percentage of cases completed without workflow assistance.

- Frequency of manual overrides by classification type.
- User-reported confidence and ease-of-use score.
- Number of review/export failures per 100 cases.

Training and support expectations

- Short onboarding session with 2-3 example cases.
- Quick-reference guide for labels, events, and common errors.
- Clear escalation path for suspected rule issues.
- Versioned release notes when rule logic or labels change.

1. Document Control

- Version: 0.2 (Expanded Draft)
- Status: Working Draft
- Last Updated: 2026-04-15
- Audience: Product, Engineering, QA, Supervisors, Clinical Advisors, and Contributors
- Repository: Open-source project repository (to be finalized)

2. PRD-Style Introduction

2.1 Vision

Build an open-source web tool that helps users analyze and annotate cardiotocography (CTG) data in a transparent, clinically aligned, and user-friendly way. The tool should assist interpretation by automatically applying rule-based annotations aligned to the National Clinical Practice Guideline: Fetal Heart Rate Monitoring (Royal College of Physicians of Ireland, RCPI).

2.2 Product Goal

Deliver a practical v1 platform for historical CTG trace review that supports:

- Importing CTG datasets.
- Interactive waveform visualization.
- Automatic rule-based annotation.
- Manual review and override workflow.
- Export of findings and audit history.

2.3 Problem Statement

CTG interpretation can be time-consuming and variable between reviewers. Teams need a consistent, explainable, and reusable tool for education, retrospective review, and algorithm development. Existing tooling may be closed, expensive, or not designed for transparent rule-level validation.

2.4 Product Principles

- Clinical decision support only (not a diagnostic replacement).
- Explainability first (show why classifications are produced).
- Human-centered design (intuitive workflow, low training overhead).
- Safety and traceability (versioned rules, reproducible outputs, audit trail).
- Open development (clear contribution and governance model).

2.5 Release Roadmap

- v1 (current scope): Historical CTG upload, visualization, rule-based annotation, manual review, export.
- v2 (future): Optional near-real-time stream ingestion and live review dashboards.

- v3 (future/research): ML-assisted risk scoring and prediction for live cases, validated in research mode before any clinical claims.

3. Scope

3.1 In Scope (v1)

- Historical CTG data ingestion from approved formats.
- Fetal heart rate (FHR) and uterine activity visualization.
- Automatic annotation based on RCPI-aligned rule set.
- Classification into Normal, Suspicious, Pathological.
- Traffic-light severity visuals with continuous gradient intensity.
- Manual annotation editing with rationale and audit logging.
- Export of annotated reports and machine-readable outputs.

3.2 Out of Scope (v1)

- Autonomous diagnosis or treatment recommendation.
- Direct integration into hospital clinical systems by default.
- Production-grade live bedside monitoring workflows.
- ML-based prediction as a primary classifier.

4. Stakeholders and Users

4.1 Stakeholders

- Project Supervisor/Product Owner
- Intern Developer (primary implementer)
- Clinical Advisor/Reviewer
- QA/Validation Reviewer
- Open-source Contributors

4.2 User Personas

- Clinical reviewer: needs readable traces, clear classification, and edit capability.
- Research user: needs reproducible annotations and exportable data.
- Intern/developer user: needs modular code, testability, and clear requirements.

5. Safety and Compliance Position

- The tool is a decision-support aid and educational/research platform.
- Final clinical interpretation and responsibility remain with qualified clinicians.
- Every UI surface that presents classifications must include a concise safety disclaimer.
- Clinical logic references RCPI guideline criteria and must be traceable.

6. Constraints and Assumptions

- Development is intern-led; requirements must be explicit and testable.
- The tool must remain open-source friendly and easy to run locally.
- Clinical rule updates require supervisor and clinical advisor sign-off.
- Data used in development/testing should be de-identified when containing patient information.

7. Requirement Priority Model

- Must: required for v1 release.
- Should: strongly preferred for v1.x if feasible.
- Could: optional enhancements that do not block release.

8. Functional Requirements (FR)

8.1 Data Ingestion

- FR-001 (Must): System shall import CTG data files in approved formats (initial format list defined in implementation notes).
- FR-002 (Must): System shall validate file structure and show user-friendly errors for invalid uploads.
- FR-003 (Should): System should support batch import of multiple traces.

8.2 Visualization

- FR-004 (Must): System shall render synchronized time-series charts for FHR and uterine activity.
- FR-005 (Must): System shall provide zoom, pan, and time-window selection.
- FR-006 (Must): System shall support overlay of detected events and annotations on the chart.
- FR-007 (Should): System should provide configurable display presets (clinical readability mode, compact mode).

8.3 Annotation Workflow

- FR-008 (Must): System shall run automatic annotation on selected or full trace windows.
- FR-009 (Must): System shall classify each analysis window as Normal, Suspicious, or Pathological.
- FR-010 (Must): System shall allow manual override (accept/edit/reject) of auto-annotations.
- FR-011 (Must): Manual overrides shall require optional rationale and store author + timestamp.
- FR-012 (Should): System should support annotation comments and bookmarks.

8.4 Reporting and Export

- FR-013 (Must): System shall export annotated results in at least one human-readable format and one machine-readable format.
- FR-014 (Must): Exports shall include rule version, analysis timestamp, and classification summary.
- FR-015 (Should): System should export a per-window event log and override history.

9. Clinical Rule Engine Requirements (CR)

9.1 Rule Alignment

- CR-001 (Must): Rule engine shall implement RCPI-aligned criteria for core CTG feature interpretation.
- CR-002 (Must): Each implemented rule shall map to a documented guideline reference in a traceability matrix.
- CR-003 (Must): Rule definitions shall be versioned independently of application release version.

9.2 Core Feature Computation

- CR-004 (Must): System shall compute baseline fetal heart rate.
- CR-005 (Must): System shall compute variability metrics.
- CR-006 (Must): System shall detect accelerations.
- CR-007 (Must): System shall detect decelerations with available subtype/severity context where data permits.
- CR-008 (Must): System shall compute contraction context/frequency from uterine activity.

9.3 Classification and Explainability

- CR-009 (Must): System shall generate per-window classification: Normal, Suspicious, Pathological.
- CR-010 (Must): System shall present explanation of triggered rules for each classification.
- CR-011 (Must): Explanations shall include threshold/evidence values used in the decision.

- CR-012 (Should): System should provide a confidence indicator derived from rule consistency/ completeness (not ML confidence).

9.4 Auditability

- CR-013 (Must): System shall store analysis metadata: input segment reference, rule version, timestamp.
- CR-014 (Must): System shall maintain immutable history of auto-generated and manually edited annotations.
- CR-015 (Must): Re-running the same input with same rule version shall produce reproducible outputs.

10. Traffic-Light Severity Visualization Requirements (VIS)

- VIS-001 (Must): System shall use traffic-light semantic classes:
  - Green: Normal
  - Orange: Suspicious
  - Red: Pathological
- VIS-002 (Must): Chart and related data views shall use a continuous severity gradient within and across classes.
- VIS-003 (Must): UI shall not rely on color alone; labels/icons/patterns shall encode class state.
- VIS-004 (Must): Colors and text contrast shall meet WCAG AA minimum contrast targets.
- VIS-005 (Should): System should provide colorblind-safe alternate palette mode.
- VIS-006 (Must): Hover/selection shall reveal exact severity score, class label, and rule explanation.

11. HCI and UX Requirements (UX)

- UX-001 (Must): Primary workflow shall follow consistent steps: Upload -> Analyze -> Review -> Export.
- UX-002 (Must): First-time users shall be able to complete a basic analysis workflow in 10 minutes or less with built-in guidance.
- UX-003 (Must): UI language shall be clear, actionable, and non-technical where possible.
- UX-004 (Must): Errors shall provide cause + suggested resolution.
- UX-005 (Must): Interface shall be desktop-first for analysis and responsive for tablet/mobile review.
- UX-006 (Must): Keyboard accessibility and meaningful labels shall be provided for key workflows.
- UX-007 (Should): Offer a clinical readability mode with larger text and simplified controls.

12. Non-Functional Requirements (NFR)

12.1 Performance

- NFR-001 (Must): For reference dataset size defined in test plan, chart rendering and interaction shall remain responsive.
- NFR-002 (Must): Annotation pipeline shall complete within acceptable target time defined by benchmark cases.

12.2 Reliability and Maintainability

- NFR-003 (Must): System shall handle malformed inputs gracefully without crash.
- NFR-004 (Must): Codebase shall be modular (ingestion, rules, visualization, export).
- NFR-005 (Must): Public functions in core logic shall include tests and concise docs.

12.3 Security and Privacy

- NFR-006 (Must): Development and sample pipelines shall assume de-identified patient data.

- NFR-007 (Must): If user identity is implemented, role separation for reviewer/editor actions shall be documented.
- NFR-008 (Should): Exported files should include configurable anonymization options.

13. QA and Validation Requirements (QA)

- QA-001 (Must): Unit tests shall validate each implemented RCPI-aligned rule.
- QA-002 (Must): Integration tests shall cover end-to-end flow: import -> analyze -> annotate -> export.
- QA-003 (Must): Visual regression tests shall cover severity gradient and class overlays.
- QA-004 (Must): Benchmark dataset with expected labels shall be used for release validation.
- QA-005 (Must): Validation report shall include per-class metrics and known limitations.
- QA-006 (Should): Usability test with at least 3 representative users before v1 release.

14. Open-Source and Governance Requirements (OSS)

- OSS-001 (Must): Repository shall include license, contribution guide, and code of conduct.
- OSS-002 (Must): Issue templates shall separate bug reports, feature requests, and clinical-rule change requests.
- OSS-003 (Must): Pull requests touching clinical logic shall require supervisor review.
- OSS-004 (Should): Architecture decision records should document major technology choices.

15. Intern Delivery Plan and Milestones

Milestone 1: Foundations

- Define data model, import pipeline, and baseline project architecture.
- Deliverable: running app skeleton + sample dataset load.

Milestone 2: Visualization and Manual Annotation

- Implement waveform charting, time controls, and manual annotation tools.
- Deliverable: reviewer can annotate and export manually.

Milestone 3: Auto-Annotation Rule Engine

- Implement RCPI-aligned rules, explanations, and class outputs.
- Deliverable: reproducible auto-annotations with traceable rule mapping.

Milestone 4: QA, Documentation, and Release Candidate

- Complete test suite, benchmark validation, and user documentation.
- Deliverable: v1 release candidate with known limitations list.

16. Implementation Freedom Boundaries

- Intern may choose framework/library stack if it satisfies maintainability, testability, and documentation criteria.
- Intern may propose better UX or data structures if requirements and safety constraints remain satisfied.
- Intern must not alter clinical classification definitions or thresholds without sign-off.
- Any rule interpretation ambiguity must be recorded and escalated for clinical review.

17. Definition of Done (Per Feature)

A feature is complete only when all checks pass:

- DoD-001: Requirement ID(s) linked in implementation notes.
- DoD-002: Automated tests added/updated and passing.
- DoD-003: UX acceptance criteria met (including accessibility where applicable).

- DoD-004: User-facing docs updated.
- DoD-005: For clinical logic, rule mapping and validation evidence updated.

18. Risks and Mitigations

- Risk: Misinterpretation of guideline language.
  - Mitigation: traceability matrix + clinical advisor review checkpoints.
- Risk: Overpromising ML capability too early.
  - Mitigation: keep ML in explicit future/research scope.
- Risk: Intern overwhelm due to unclear priorities.
  - Mitigation: Must/Should/Could prioritization and milestone-based delivery.
- Risk: UI misuse due to color-only encoding.
  - Mitigation: mandatory multi-channel severity indicators.

19. Future Considerations (Non-v1)

- Live stream ingestion adapters for near-real-time CTG display.
- Event-driven alerting for sustained pathological patterns.
- ML-assisted triage models with transparent evaluation and bias analysis.
- Integration adapters for clinical information systems after governance review.

20. Approval and Change Control

- This SRS is a living document.
- Any change to Must requirements or clinical rules requires documented review and sign-off.
- Change log entries should include reason, owner, and effective version.

21. Appendix A - RCPI Rule Traceability Matrix Template

Use this template to map each implemented clinical rule to the RCPI guideline source and the validating test case.

21.1 Rule Definition and Mapping

Matrix ID	RCPI Clause Ref	Guideline Statement Summary	Local Rule ID	Rule Name
RTM-001	RCPI SECTION X.Y	Short rule interpretation	CR-00X-R01	Rule label

21.2 Logic and Classification Impact

Inputs Used	Decision Logic Summary	Output Class Impact	Edge Cases/ Exclusions	Implementation Module
FHR and variability inputs	Threshold and condition summary	Normal / Suspicious / Pathological	Known limitations	Module or file name

21.3 Validation and Review Status

Unit Test ID	Integration Test ID	Clinical Reviewer	Status	Notes
UT ID	IT ID	Reviewer name	Draft / Approved / Deprecated	Comments

Required completion guidance:

- One matrix row per implemented rule revision.
- `RCPI Clause Ref` must be precise enough for independent review.
- `Guideline Statement Summary` should be plain language and implementation-oriented.
- `Decision Logic Summary` must include key thresholds and comparison logic.
- `Status` transitions require documented reviewer sign-off.

22. Appendix B - Sample Benchmark Validation Report Template (QA)

Use this report format for release-gating validation on benchmark datasets.

22.1 Report Header

- Report ID: QA-VAL-YYYYMMDD-NN
- SRS Version: EXAMPLE_0_2
- App Version: EXAMPLE_V1_0_0_RC1
- Rule Set Version: EXAMPLE_RULES_2026_04
- Prepared By: REVIEWER_NAME
- Reviewed By: SUPERVISOR_NAME
- Date: YYYY-MM-DD

22.2 Objective and Scope

- Objective: Verify rule-engine and end-to-end annotation performance on benchmark CTG data.
- Scope: DATASETS_WORKFLOWS_AND_EXCLUSIONS

22.3 Benchmark Dataset Description

- Dataset Name(s): DATASET_IDENTIFIER
- Source: PUBLIC_OR_INTERNAL_AND_DEIDENT_STATUS
- Case Count: N
- Duration Distribution: DURATION_SUMMARY
- Label Distribution: NORMAL_SUSPICIOUS_PATHOLOGICAL_COUNTS
- Inclusion/Exclusion Criteria: INCLUSION_EXCLUSION_CRITERIA

22.4 Test Environment

- Runtime/Platform: OS_BROWSER_HARDWARE
- Build/Commit: COMMIT_HASH_OR_TAG
- Configuration: IMPORTANT_FLAGS_AND_THRESHOLDS
- Reproducibility Artifacts: SEED_CONFIG_AND_SCRIPTS

22.5 Evaluation Method

- Ground Truth Definition: LABELERS_AND_CONSENSUS_METHOD
- Matching Strategy: WINDOW_ALIGNMENT_AND_TOLERANCE
- Metrics:
  - Per-class precision, recall, F1
  - Confusion matrix
  - Overall accuracy (if appropriate)
  - Processing time per trace/window
- Statistical Notes: CONFIDENCE_INTERVALS_OR_CAVEATS

22.6 Results Summary (Sample Table)

Metric	Normal	Suspicious	Pathological	Overall
Precision	VALUE	VALUE	VALUE	VALUE
Recall	VALUE	VALUE	VALUE	VALUE
F1 Score	VALUE	VALUE	VALUE	VALUE
Support (n)	VALUE	VALUE	VALUE	TOTAL

22.7 Confusion Matrix (Sample)

Actual / Predicted	Normal	Suspicious	Pathological

Normal	VALUE	VALUE	VALUE
Suspicious	VALUE	VALUE	VALUE
Pathological	VALUE	VALUE	VALUE

22.8 Rule-Level Findings

Rule ID	Cases Triggered	Agreement with Expected	Common Failure Pattern	Severity	Action Required
CR-00X-R01	N	Percent	Short note	Low / Medium / High	Task reference

22.9 Usability and Visualization Checks

- Traffic-light gradient visible and consistent across chart and table views.
- Color-independent encoding present (labels/icons/patterns).
- Accessibility spot-check completed (contrast and keyboard flow).

22.10 Known Limitations

- LIMITATION_1
- LIMITATION_2
- LIMITATION_3

22.11 Release Decision

- Decision: Pass / Conditional Pass / Fail
- Blocking Issues: ISSUE_IDS
- Required Follow-up Actions: ACTIONS_OWNERS_AND_DUE_DATES